An Information Theory of Finite Abstractions: fundamental scalability limits and what to do next

Giannis Delimpaltadakis, Gabriel Gleizer







Complex dynamics with complex specifications



How to guarantee complex specifications?

Finite Abstractions

Finite models that include all behaviors of the original system.

- Computationally handleable.
- Formal guarantees

Finite Abstractions

Finite models that include all behaviors of the original system.

- Computationally handleable.
- Formal guarantees (abstraction safe $\implies$ system safe).

Finite Abstractions

Finite models that include all behaviors of the original system.

- Computationally handleable.
- Formal guarantees (abstraction safe $\implies$ system safe).
- Partition state (and control) space.

Three decades of research... and the curse of dimensionality

- System classes (nonlinear, hybrid, stochastic, data-driven...).
- Abstraction classes (automata, MDPs).
- Types of specifications (safety, reach-avoid, LTL, STL...).

Three decades of research... and the curse of dimensionality

- System classes (nonlinear, hybrid, stochastic, data-driven...).
- Abstraction classes (automata, MDPs).
- Types of specifications (safety, reach-avoid, LTL, STL...).

A 30-y.o. consensus:

Theorem (Not quite a theorem)

Abstractions do not scale.

- *For better accuracy, partition more.*
- *For the same accuracy, partition size grows exponentially with system dimensions.*

Three decades of research... and the curse of dimensionality

A 30-y.o. consensus:

Theorem (Not quite a theorem)

Abstractions do not scale.

- *For better accuracy, partition more.*
- *For the same accuracy, partition size grows exponentially with system dimensions.*

No formal theory on size-accuracy tradeoff:

Three decades of research... and the curse of dimensionality

A 30-y.o. consensus:

Theorem (Not quite a theorem)

Abstractions do not scale.

- *For better accuracy, partition more.*
- *For the same accuracy, partition size grows exponentially with system dimensions.*

No formal theory on size-accuracy tradeoff:

- Scalability limits unknown.
  - Given a system and a desired accuracy, how big should the abstraction be?

Three decades of research... and the curse of dimensionality

A 30-y.o. consensus:

Theorem (Not quite a theorem)

Abstractions do not scale.

- *For better accuracy, partition more.*
- *For the same accuracy, partition size grows exponentially with system dimensions.*

No formal theory on size-accuracy tradeoff:

- Scalability limits unknown.
  - Given a system and a desired accuracy, how big should the abstraction be?
- Heuristic methods towards scalable abstractions.

Contribution

Main Contribution

A statistical theory on abstractions' size-accuracy tradeoff and scalability limits.

Contribution

Main Contribution

A statistical theory on abstractions' size-accuracy tradeoff and scalability limits.

- Connection to rate-distortion theory.
  - Abstractions are encoders of system trajectories.
  - Distortion (accuracy): spatial average deviation between abstract and system trajectories.

Contribution

Main Contribution

A statistical theory on abstractions' size-accuracy tradeoff and scalability limits.

- Connection to rate-distortion theory.
  - Abstractions are encoders of system trajectories.
  - Distortion (accuracy): spatial average deviation between abstract and system trajectories.
- Fundamental scalability limits.
  - Lower bound on abstraction size, given system dynamics and desired accuracy.
  - Lower bound on distortion, given dynamics and abstraction size.

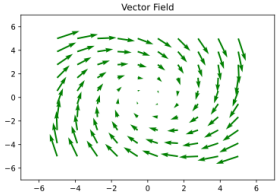
Contribution

Main Contribution

A statistical theory on abstractions' size-accuracy tradeoff and scalability limits.

- Connection to rate-distortion theory.
  - Abstractions are encoders of system trajectories.
  - Distortion (accuracy): spatial average deviation between abstract and system trajectories.
- Fundamental scalability limits.
  - Lower bound on abstraction size, given system dynamics and desired accuracy.
  - Lower bound on distortion, given dynamics and abstraction size.
- Enables minimal abstractions.

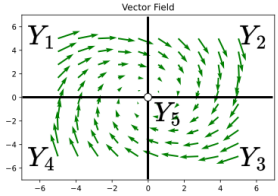
Abstractions



System S:

- $x(k+1) = f(x(k))$, $x \in \mathcal{X} \subset \mathbb{R}^n$ compact.

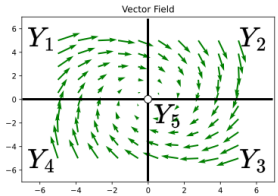
Abstractions

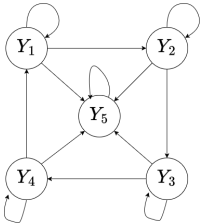


System S:

- $x(k+1) = f(x(k))$, $x \in \mathcal{X} \subset \mathbb{R}^n$ compact.

Abstractions



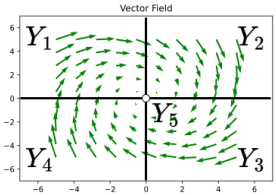


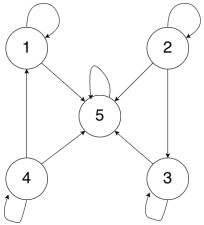
Abstraction

System S:

- $x(k+1) = f(x(k))$, $x \in \mathcal{X} \subset \mathbb{R}^n$ compact.

Abstractions



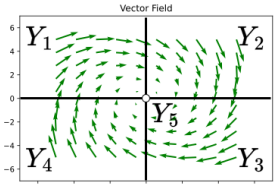


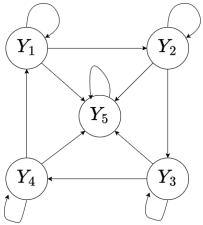
Not an abstraction!

System S:

- $x(k+1) = f(x(k))$, $x \in \mathcal{X} \subset \mathbb{R}^n$ compact.

Abstractions





System S:

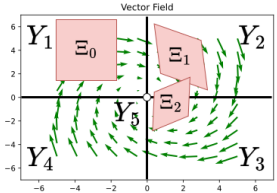
- $x(k+1) = f(x(k))$, $x \in \mathcal{X} \subset \mathbb{R}^n$ compact.

Abstraction $A = (\mathcal{Y}, \mathrm{Tran}_A)$:

- Partition: $\mathcal{Y} = \{Y_i\}$.
- $\mathrm{Tran}_A := \Big\{(Y_1, Y_1),\ (Y_1, Y_2), (Y_1, Y_5), \dots\Big\}$.

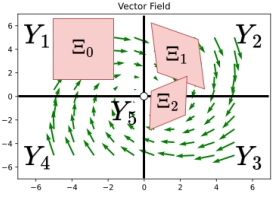
Abstractions - Verification

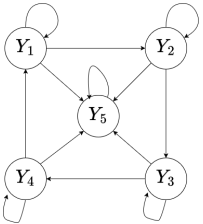
- Initial condition set: $\Xi_0 \subseteq \mathcal{X}$.
- Actual reachable set: $\Xi = \Xi_0 \times \Xi_1 \times \cdots \times \Xi_{l-1} \subseteq \mathbb{R}^{nl}$
- Abstract reachable set $\Omega_A \supseteq \Xi$.



Abstractions - Verification

- Initial condition set: $\Xi_0 \subseteq \mathcal{X}$.
- Actual reachable set: $\Xi = \Xi_0 \times \Xi_1 \times \cdots \times \Xi_{l-1} \subseteq \mathbb{R}^{nl}$
- Abstract reachable set $\Omega_A \supseteq \Xi$.

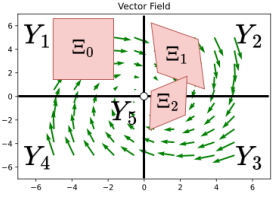


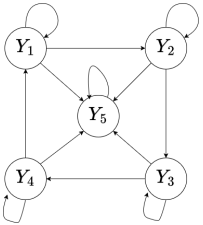


$$\Omega_A = \underbrace{Y_1}_{\Xi_0 \subseteq Y_1} \times \underbrace{(Y_1 \cup Y_2 \cup Y_5)}_{\Xi_1 \subseteq Y_1 \cup Y_2 \cup Y_5} \times \underbrace{(Y_1 \cup Y_2 \cup Y_3 \cup Y_5)}_{\Xi_2 \subseteq Y_1 \cup Y_2 \cup Y_3 \cup Y_5}$$

Abstractions - Verification

- Initial condition set: $\Xi_0 \subseteq \mathcal{X}$.
- Actual reachable set: $\Xi = \Xi_0 \times \Xi_1 \times \cdots \times \Xi_{l-1} \subseteq \mathbb{R}^{nl}$
- Abstract reachable set $\Omega_A \supseteq \Xi$.





$$\Omega_A = \underbrace{Y_1}_{\Xi_0 \subseteq Y_1} \times \underbrace{(Y_1 \cup Y_2 \cup Y_5)}_{\Xi_1 \subseteq Y_1 \cup Y_2 \cup Y_5} \times \underbrace{(Y_1 \cup Y_2 \cup Y_3 \cup Y_5)}_{\Xi_2 \subseteq Y_1 \cup Y_2 \cup Y_3 \cup Y_5}$$

- Ω_A much bigger than Ξ!
- Size-accuracy tradeoff.
- Optimality?

Abstractions - Behavioral inclusion

System l-long behavior:

$$\underbrace{\mathcal{B}_l^S \coloneqq \left\{ \{\xi_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \ \xi_0 \in \mathcal{X}, \ \xi_i = f(\xi_{i-1}) \right\}}_{n\text{-dimensional}}.$$

Abstraction l-long behavior:

$$\underbrace{\mathcal{B}_l^A \coloneqq \left\{ \{\omega_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \omega_0 \in \mathcal{Y}, (\omega_i, \omega_{i+1}) \in \mathrm{Tran}_A \right\}}_{nl\text{-dimensional}}.$$

Abstractions - Behavioral inclusion

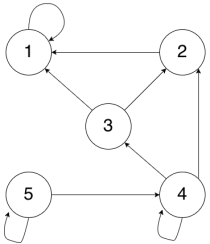
System l-long behavior:

$$\underbrace{\mathcal{B}_l^S \coloneqq \left\{\{\xi_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \ \xi_0 \in \mathcal{X}, \ \xi_i = f(\xi_{i-1})\right\}}_{n\text{-dimensional}}.$$

Abstraction l-long behavior:

$$\underbrace{\mathcal{B}_l^A \coloneqq \left\{\{\omega_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \omega_0 \in \mathcal{Y}, (\omega_i, \omega_{i+1}) \in \mathrm{Tran}_A\right\}}_{nl\text{-dimensional}}.$$

- $x(k+1) = x(k)^2$, $\mathcal{X} = [0,1]$.
- $\mathcal{Y} = \{[0,0.2), [0.2,0.4), \ldots, [0.8,1]\}$.



Abstractions - Behavioral inclusion

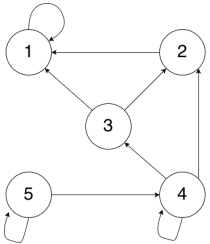
System l-long behavior:

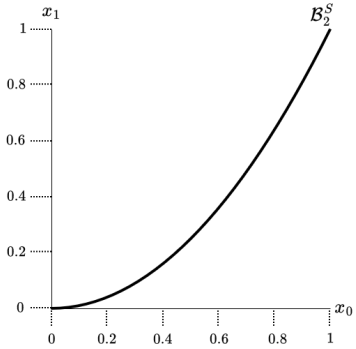
$$\underbrace{\mathcal{B}_l^S \coloneqq \left\{\{\xi_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \ \xi_0 \in \mathcal{X},\ \xi_i = f(\xi_{i-1})\right\}}_{n\text{-dimensional}}.$$

Abstraction l-long behavior:

$$\underbrace{\mathcal{B}_l^A \coloneqq \left\{\{\omega_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \omega_0 \in \mathcal{Y}, (\omega_i, \omega_{i+1}) \in \mathrm{Tran}_A\right\}}_{nl\text{-dimensional}}.$$

- $x(k+1) = x(k)^2$, $\mathcal{X} = [0,1]$.
- $\mathcal{Y} = \{[0,0.2), [0.2,0.4), \ldots, [0.8,1]\}$.





Abstractions - Behavioral inclusion

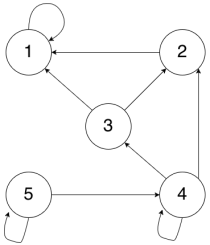
System l-long behavior:

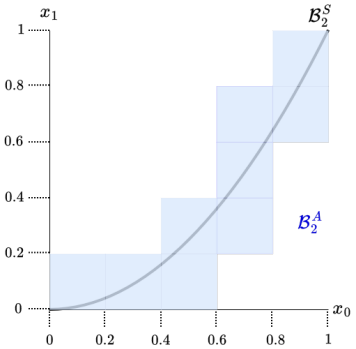
$$\underbrace{\mathcal{B}_l^S \coloneqq \left\{\{\xi_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \ \xi_0 \in \mathcal{X},\ \xi_i = f(\xi_{i-1})\right\}}_{n\text{-dimensional}}.$$

Abstraction l-long behavior:

$$\underbrace{\mathcal{B}_l^A \coloneqq \left\{\{\omega_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \omega_0 \in \mathcal{Y}, (\omega_i, \omega_{i+1}) \in \mathrm{Tran}_A\right\}}_{nl\text{-dimensional}}.$$

- $x(k+1) = x(k)^2$, $\mathcal{X} = [0,1]$.
- $\mathcal{Y} = \{[0,0.2), [0.2,0.4), \ldots, [0.8,1]\}$.





Abstractions - Behavioral inclusion

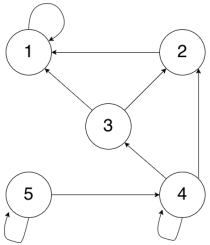
System l-long behavior:

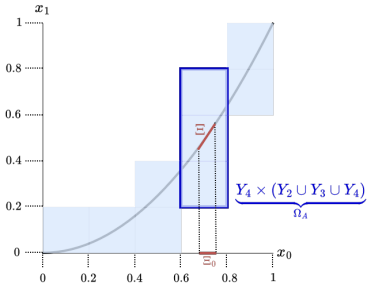
$$\underbrace{\mathcal{B}_l^S \coloneqq \left\{\{\xi_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \ \xi_0 \in \mathcal{X},\ \xi_i = f(\xi_{i-1})\right\}}_{n\text{-dimensional}}.$$

Abstraction l-long behavior:

$$\underbrace{\mathcal{B}_l^A \coloneqq \left\{\{\omega_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \omega_0 \in \mathcal{Y}, (\omega_i, \omega_{i+1}) \in \mathrm{Tran}_A\right\}}_{nl\text{-dimensional}}.$$

- $x(k+1) = x(k)^2$, $\mathcal{X} = [0,1]$.
- $\mathcal{Y} = \{[0, 0.2), [0.2, 0.4), \ldots, [0.8, 1]\}$.





Abstractions - Behavioral inclusion

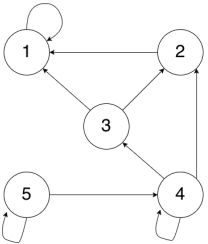
System l-long behavior:

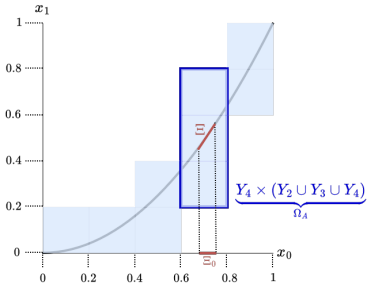
$$\underbrace{\mathcal{B}_l^S \coloneqq \left\{\{\xi_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \ \xi_0 \in \mathcal{X}, \ \xi_i = f(\xi_{i-1})\right\}}_{n\text{-dimensional}}.$$

Abstraction l-long behavior:

$$\underbrace{\mathcal{B}_l^A \coloneqq \left\{\{\omega_i\}_{i=0}^{l-1} \in \mathbb{R}^{nl} : \omega_0 \in \mathcal{Y}, (\omega_i, \omega_{i+1}) \in \mathrm{Tran}_A\right\}}_{nl\text{-dimensional}}.$$

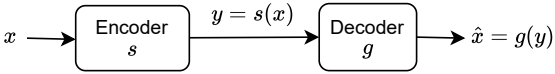
- $x(k+1) = x(k)^2$, $\mathcal{X} = [0,1]$.
- $\mathcal{Y} = \{[0,0.2), [0.2,0.4), \ldots, [0.8,1]\}$.





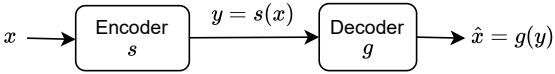
$\mathcal{B}_l^A$ is a finite nl-dimensional cover of $\mathcal{B}_l^S$

Rate-Distortion theory



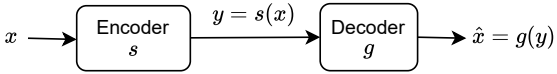
- Message $x \in \mathcal{X} \subseteq \mathbb{R}^n$, with $p_x : \mathcal{X} \to \mathbb{R}_+$.
- Encoder $s : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y}$ finite.
- Rate $\log |\mathcal{Y}|$.

Rate-Distortion theory



- Message $x \in \mathcal{X} \subseteq \mathbb{R}^n$, with $p_x : \mathcal{X} \to \mathbb{R}_+$.
- Encoder $s : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y}$ finite.
- Rate $\log |\mathcal{Y}|$.
- Decoder $g : \mathcal{Y} \to \hat{\mathcal{X}} \subseteq \mathbb{R}^n$.
- Distortion $d : \mathcal{X} \times \hat{\mathcal{X}} \to \mathbb{R}_+$ (e.g. $d(x, \hat{x}) = \|x - \hat{x}\|^2$).

Rate-Distortion theory

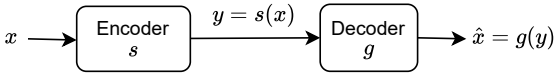


- Message $x \in \mathcal{X} \subseteq \mathbb{R}^n$, with $p_x : \mathcal{X} \to \mathbb{R}_+$.
- Encoder $s : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y}$ finite.
- Rate $\log |\mathcal{Y}|$.
- Decoder $g : \mathcal{Y} \to \hat{\mathcal{X}} \subseteq \mathbb{R}^n$.
- Distortion $d : \mathcal{X} \times \hat{\mathcal{X}} \to \mathbb{R}_+$ (e.g. $d(x, \hat{x}) = \|x - \hat{x}\|^2$).

Optimal Rate-Distortion tradeoff

$$D(R) := \inf_{s,g} \mathbb{E}_x[d(x, \hat{x}) \mid s, g], \quad \text{s.t. } \log |\mathcal{Y}| \leq R, \ \hat{x} = g(s(x))$$

Rate-Distortion theory



- Message $x \in \mathcal{X} \subseteq \mathbb{R}^n$, with $p_x : \mathcal{X} \to \mathbb{R}_+$.
- Encoder $s : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y}$ finite.
- Rate $\log |\mathcal{Y}|$.
- Decoder $g : \mathcal{Y} \to \hat{\mathcal{X}} \subseteq \mathbb{R}^n$.
- Distortion $d : \mathcal{X} \times \hat{\mathcal{X}} \to \mathbb{R}_+$ (e.g. $d(x, \hat{x}) = \|x - \hat{x}\|^2$).

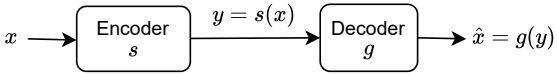
Optimal Rate-Distortion tradeoff

$$D(R) := \inf_{s,g} \mathbb{E}_x[d(x, \hat{x}) \mid s, g], \quad \text{s.t. } \log |\mathcal{Y}| \leq R, \ \hat{x} = g(s(x))$$

Theorem (Generalized Shannon Lower Bound (SLB))

Assume $\mathcal{X}$ is m-dimensional and $d(x, \hat{x}) = \|x - \hat{x}\|^2$. Then $D(R) \geq \frac{m}{2} \left(\frac{e^{-R + h(x) - m/2}}{c_{\mathcal{X}} \Gamma(1 + m/2)} \right)^{2/m}$.

Rate-Distortion theory



- Message $x \in \mathcal{X} \subseteq \mathbb{R}^n$, with $p_x : \mathcal{X} \to \mathbb{R}_+$.
- Encoder $s : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y}$ finite.
- Rate $\log |\mathcal{Y}|$.
- Decoder $g : \mathcal{Y} \to \hat{\mathcal{X}} \subseteq \mathbb{R}^n$.
- Distortion $d : \mathcal{X} \times \hat{\mathcal{X}} \to \mathbb{R}_+$ (e.g. $d(x, \hat{x}) = \|x - \hat{x}\|^2$).

- Entropy $h(x)$.
- Geometric constant $c_{\mathcal{X}}$.
- Gamma function Γ.

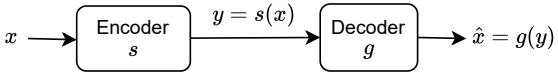
Optimal Rate-Distortion tradeoff

$$D(R) := \inf_{s,g} \mathbb{E}_x[d(x, \hat{x}) \mid s, g], \quad \text{s.t. } \log |\mathcal{Y}| \le R, \ \hat{x} = g(s(x))$$

Theorem (Generalized Shannon Lower Bound (SLB))

Assume $\mathcal{X}$ is m-dimensional and $d(x, \hat{x}) = \|x - \hat{x}\|^2$. Then $D(R) \ge \frac{m}{2} \left(\frac{e^{-R + h(x) - m/2}}{c_{\mathcal{X}} \Gamma(1 + m/2)} \right)^{2/m}$.

Rate-Distortion theory



- Message $x \in \mathcal{X} \subseteq \mathbb{R}^n$, with $p_x : \mathcal{X} \to \mathbb{R}_+$.
- Encoder $s : \mathcal{X} \to \mathcal{Y}$, where $\mathcal{Y}$ finite.
- Rate $\log |\mathcal{Y}|$.
- Decoder $g : \mathcal{Y} \to \hat{\mathcal{X}} \subseteq \mathbb{R}^n$.
- Distortion $d : \mathcal{X} \times \hat{\mathcal{X}} \to \mathbb{R}_+$ (e.g. $d(x, \hat{x}) = \|x - \hat{x}\|^2$).

- Entropy $h(x)$.
- Geometric constant $c_{\mathcal{X}}$.
- Gamma function Γ.

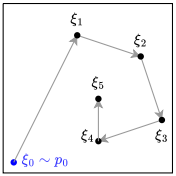
Optimal Rate-Distortion tradeoff

$$R(D) := \inf_{s,g} \log |\mathcal{Y}|, \quad \text{s.t. } \mathbb{E}_x[d(x, \hat{x}) \mid s, g] \leq D, \ \hat{x} = g(s(x))$$

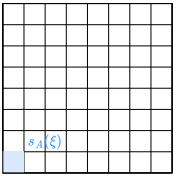
Theorem (Generalized Shannon Lower Bound (SLB))

Assume $\mathcal{X}$ is m-dimensional and $d(x, \hat{x}) = \|x - \hat{x}\|^2$. Then
$R(D) \geq h(x) - \frac{m}{2} - \log\left(\frac{c_{\mathcal{X}} D^{m/2} \Gamma(1+m/2)}{(\frac{m}{2})^{m/2}}\right)$.

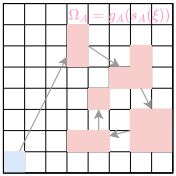
Information theory of abstractions - Encoding trajectories



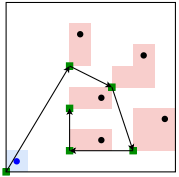
Message ξ



Encoder s_A

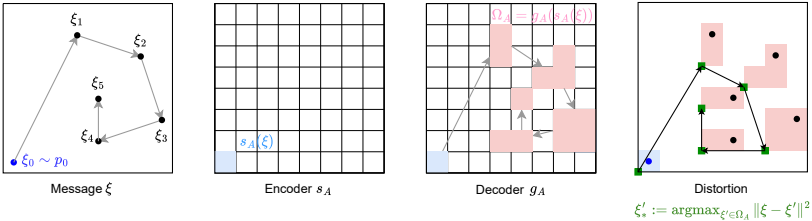


Decoder g_A


Distortion

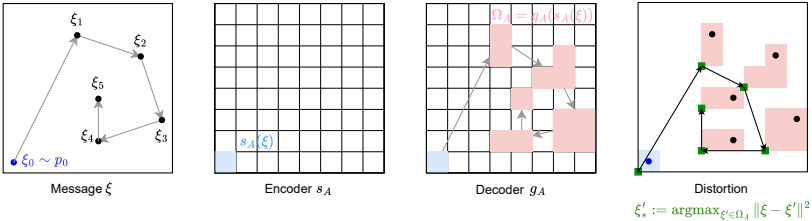
$\xi'_* := \text{argmax}_{\xi' \in \Omega_A} \|\xi - \xi'\|^2$

Information theory of abstractions - Encoding trajectories



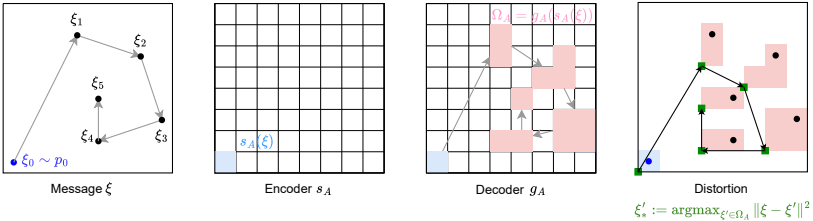
- Encoder $s_A : \mathcal{B}_I^{\mathcal{S}} \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.

Information theory of abstractions - Encoding trajectories



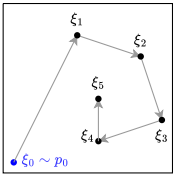
- Encoder $s_A : \mathcal{B}_l^{\mathcal{S}} \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}^l}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.

Information theory of abstractions - Encoding trajectories

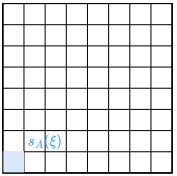


- Encoder $s_A : \mathcal{B}_l^{\mathcal{S}} \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}^l}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.
- Distortion $d(\xi, \Omega_A) \coloneqq \sup_{\xi' \in \Omega_A} \frac{1}{l} \|\xi - \xi'\|^2$, where $\Omega_A = g_A(s_A(\xi))$.

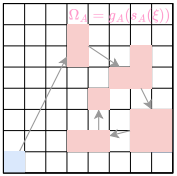
Information theory of abstractions - Encoding trajectories



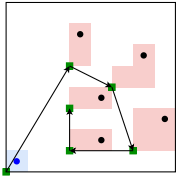
Message ξ



Encoder s_A

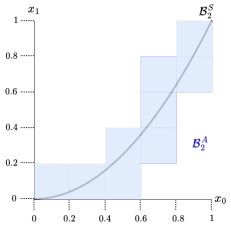


Decoder g_A


Distortion

$\xi'_* := \text{argmax}_{\xi' \in \Omega_A} \|\xi - \xi'\|^2$

- Encoder $s_A : \mathcal{B}_l^S \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}^l}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.
- Distortion $d(\xi, \Omega_A) \coloneqq \sup_{\xi' \in \Omega_A} \frac{1}{l} \|\xi - \xi'\|^2$, where $\Omega_A = g_A(s_A(\xi))$.



Abstractions are encoder-decoder pairs that encode and cover $\mathcal{B}_l^S$.

Information theory of abstractions - Size-Accuracy tradeoff

- Encoder $s_A : \mathcal{B}_l^S \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}^l}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.
- Distortion $d(\xi, \Omega_A) \coloneqq \sup_{\xi' \in \Omega_A} \frac{1}{l} \|\xi - \xi'\|^2$, where $\Omega_A = g_A(s_A(\xi))$.

Information theory of abstractions - Size-Accuracy tradeoff

- Encoder $s_A : \mathcal{B}_l^S \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}^l}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.
- Distortion $d(\xi, \Omega_A) \coloneqq \sup_{\xi' \in \Omega_A} \frac{1}{l} \|\xi - \xi'\|^2$, where $\Omega_A = g_A(s_A(\xi))$.

Abstraction Size - Rate

Rate $= \log |\mathcal{Y}|$.

Information theory of abstractions - Size-Accuracy tradeoff

- Encoder $s_A : \mathcal{B}_l^S \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}'}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.
- Distortion $d(\xi, \Omega_A) \coloneqq \sup_{\xi' \in \Omega_A} \frac{1}{l} \|\xi - \xi'\|^2$, where $\Omega_A = g_A(s_A(\xi))$.

Abstraction Size - Rate

Rate $= \log |\mathcal{Y}|$.

- Initial condition $\xi_0 \sim p_{\xi_0}$ (implies $\xi \sim p_\xi$).

Information theory of abstractions - Size-Accuracy tradeoff

- Encoder $s_A : \mathcal{B}_l^{\mathcal{S}} \to \mathcal{Y}$, with $s_A(\xi) \coloneqq Y$, where $\xi_0 \in Y$.
- Decoder $g_A : \mathcal{Y} \to 2^{\mathcal{X}'}$, with $g_A(y) \coloneqq \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega$.
- Distortion $d(\xi, \Omega_A) \coloneqq \sup_{\xi' \in \Omega_A} \frac{1}{l} \|\xi - \xi'\|^2$, where $\Omega_A = g_A(s_A(\xi))$.

Abstraction Size - Rate

Rate $= \log |\mathcal{Y}|$.

- Initial condition $\xi_0 \sim p_{\xi_0}$ (implies $\xi \sim p_\xi$).

Abstraction Accuracy - Average distortion

The spatial average deviation between abstract and system trajectories: $\mathbb{E}_{\xi_0}[d(\xi, \Omega_A)|A]$

Information theory of abstractions - Size-Accuracy tradeoff

Abstraction Size - Rate

Rate $= \log|\mathcal{Y}|$.

Abstraction Accuracy - Average distortion

The spatial average deviation between abstract and system trajectories: $\mathbb{E}_{\xi_0}[d(\xi, \Omega_A)|A]$

Optimal size-accuracy tradeoff

$$D_{abs}(R) := \inf_{A} \mathbb{E}_{\xi_0}[d(\xi, \Omega_A) \mid A]$$

$$\text{s.t. } A \text{ is an abstraction of given system } S,$$

$$s_A : \mathcal{B}_l^S \to \mathcal{Y}, \text{ with } s_A(\xi) := Y, \text{ where } \xi_0 \in Y,$$

$$g_A : \mathcal{Y} \to 2^{\mathcal{X}'}, \text{ with } g_A(y) := \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega,$$

$$\log|\mathcal{Y}| \le R,\ \Omega_A = g_A(s_A(\xi)).$$

Information theory of abstractions - Size-Accuracy tradeoff

Abstraction Size - Rate

Rate $= \log|\mathcal{Y}|$.

Abstraction Accuracy - Average distortion

The spatial average deviation between abstract and system trajectories: $\mathbb{E}_{\xi_0}[d(\xi, \Omega_A)|A]$

Optimal size-accuracy tradeoff

$$R_{abs}(D) := \inf_{A} \ \log|\mathcal{Y}|$$

$$\text{s.t. } A \text{ is an abstraction of given system } S,$$

$$s_A : \mathcal{B}_l^S \to \mathcal{Y}, \text{ with } s_A(\xi) := Y, \text{ where } \xi_0 \in Y,$$

$$g_A : \mathcal{Y} \to 2^{\mathcal{X}'}, \text{ with } g_A(y) := \bigcup_{\omega \in \mathcal{B}_l^A,\ \omega_0 = y} \omega,$$

$$\mathbb{E}_{\xi_0}[d(\xi, \Omega_A) \mid A] \leq D,\ \Omega_A = g_A(s_A(\xi)).$$

Information theory of abstractions - Fundamental Scalability Limits

Theorem (Fundamental limits on optimal abstraction size-accuracy tradeoff)

Assume $\mathcal{B}_l^S$ is a finite union of bounded, n-dimensional C^1-manifolds. Then:

$$D_{abs}(R) \geq \frac{n}{2l}\Big(\frac{e^{-R+h(\xi)-n/2}}{c_{\mathcal{B}_l^S}\Gamma(1+n/2)}\Big)^{2/n} + \frac{1}{l}c_{\mathcal{B}_l^S}^{-2/n} \max_{s\in(1,\infty]} e^{\frac{2}{n}(-\frac{s}{s-1}R+h_s(\xi))},$$

$$R_{abs}(D) \geq \frac{n}{2}\log\Big(\frac{n}{2lc_{\mathcal{B}_l^S}^{2/n}}\frac{e^{\frac{2}{n}h(\xi)-1}}{\Gamma(1+n/2)^{2/n}} + e^{\frac{2}{n}h_\infty(\xi)}\Big) - \frac{n}{2}\log(D).$$

Information theory of abstractions - Fundamental Scalability Limits

Theorem (Fundamental limits on optimal abstraction size-accuracy tradeoff)

Assume $\mathcal{B}_l^S$ is a finite union of bounded, n-dimensional C^1-manifolds. Then:

$$D_{abs}(R) \geq \frac{n}{2l}\Big(\frac{e^{-R+h(\xi)-n/2}}{c_{\mathcal{B}_l^S}\Gamma(1+n/2)}\Big)^{2/n} + \frac{1}{l}c_{\mathcal{B}_l^S}^{-2/n} \max_{s\in(1,\infty]} e^{\frac{2}{n}(-\frac{s}{s-1}R+h_s(\xi))},$$

$$R_{abs}(D) \geq \frac{n}{2}\log\Big(\frac{n}{2lc_{\mathcal{B}_l^S}^{2/n}}\frac{e^{\frac{2}{n}h(\xi)-1}}{\Gamma(1+n/2)^{2/n}} + e^{\frac{2}{n}h_\infty(\xi)}\Big) - \frac{n}{2}\log(D).$$

- More complex dynamics, bigger size (or bigger distortion).

Information theory of abstractions - Fundamental Scalability Limits

Theorem (Fundamental limits on optimal abstraction size-accuracy tradeoff)

Assume $\mathcal{B}_l^S$ is a finite union of bounded, n-dimensional C^1-manifolds. Then:

$$D_{abs}(R) \geq \frac{n}{2l}\Big(\frac{e^{-R+h(\xi)-n/2}}{c_{\mathcal{B}_l^S}\Gamma(1+n/2)}\Big)^{2/n} + \frac{1}{l}c_{\mathcal{B}_l^S}^{-2/n} \max_{s\in(1,\infty]} e^{\frac{2}{n}(-\frac{s}{s-1}R+h_s(\xi))},$$

$$R_{abs}(D) \geq \frac{n}{2}\log\Big(\frac{n}{2lc_{\mathcal{B}_l^S}^{2/n}}\frac{e^{\frac{2}{n}h(\xi)-1}}{\Gamma(1+n/2)^{2/n}} + e^{\frac{2}{n}h_\infty(\xi)}\Big) - \frac{n}{2}\log(D).$$

- More complex dynamics, bigger size (or bigger distortion).
- Limits are fundamental!

Information theory of abstractions - Fundamental Scalability Limits

Theorem (Fundamental limits on optimal abstraction size-accuracy tradeoff)

Assume $\mathcal{B}_l^S$ is a finite union of bounded, n-dimensional C^1-manifolds. Then:

$$D_{abs}(R) \geq \frac{n}{2l}\Big(\frac{e^{-R+h(\xi)-n/2}}{c_{\mathcal{B}_l^S}\Gamma(1+n/2)}\Big)^{2/n} + \frac{1}{l}c_{\mathcal{B}_l^S}^{-2/n}\max_{s\in(1,\infty]} e^{\frac{2}{n}(-\frac{s}{s-1}R+h_s(\xi))},$$

$$R_{abs}(D) \geq \frac{n}{2}\log\Big(\frac{n}{2lc_{\mathcal{B}_l^S}^{2/n}}\frac{e^{\frac{2}{n}h(\xi)-1}}{\Gamma(1+n/2)^{2/n}} + e^{\frac{2}{n}h_\infty(\xi)}\Big) - \frac{n}{2}\log(D).$$

- More complex dynamics, bigger size (or bigger distortion).
- Limits are fundamental!

Fundamental curse of dimensionality

$$\log|\mathcal{Y}| \geq R_{abs}(D) \geq \text{constant} - \frac{n}{2}\log(D)$$

Information theory of abstractions - Fundamental Scalability Limits

Theorem (Fundamental limits on optimal abstraction size-accuracy tradeoff)

Assume $\mathcal{B}_l^S$ is a finite union of bounded, n-dimensional C^1-manifolds. Then:

$$D_{abs}(R) \geq \frac{n}{2l}\Big(\frac{e^{-R+h(\xi)-n/2}}{c_{\mathcal{B}_l^S}\Gamma(1+n/2)}\Big)^{2/n} + \frac{1}{l}c_{\mathcal{B}_l^S}^{-2/n} \max_{s\in(1,\infty]} e^{\frac{2}{n}(-\frac{s}{s-1}R+h_s(\xi))},$$

$$R_{abs}(D) \geq \frac{n}{2}\log\Big(\frac{n}{2lc_{\mathcal{B}_l^S}^{2/n}}\frac{e^{\frac{2}{n}h(\xi)-1}}{\Gamma(1+n/2)^{2/n}} + e^{\frac{2}{n}h_\infty(\xi)}\Big) - \frac{n}{2}\log(D).$$

- More complex dynamics, bigger size (or bigger distortion).
- Limits are fundamental!

Developed rate-distortion theory for abstractions:

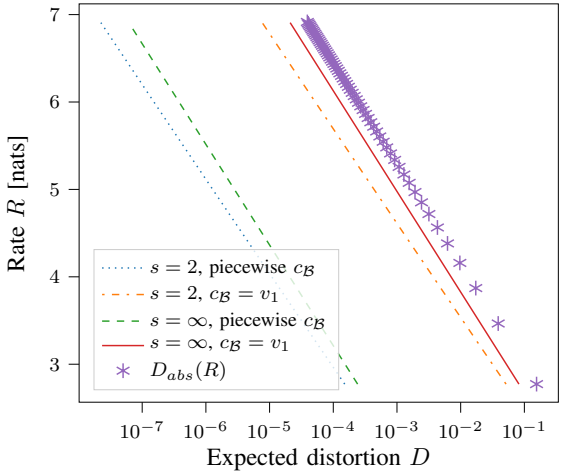
- Encoder-Decoder outputs are set-based.
- Distortion is set-based.

Chaotic dynamics - sharp bounds and a minimal abstraction

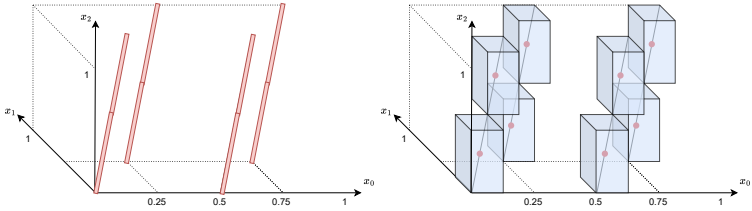
The doubling map: $x(k+1) = 2x(k) \mod 1$, $x \in \mathcal{X} = [0,1]$.

Chaotic dynamics - sharp bounds and a minimal abstraction

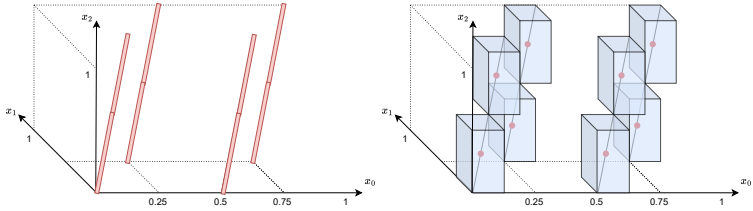
The doubling map: $x(k+1) = 2x(k) \mod 1$, $x \in \mathcal{X} = [0,1]$.



Chaotic dynamics - sharp bounds and a minimal abstraction

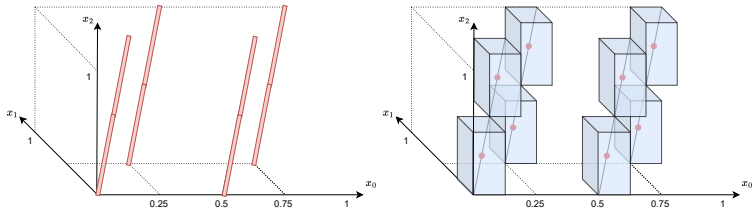


Chaotic dynamics - sharp bounds and a minimal abstraction



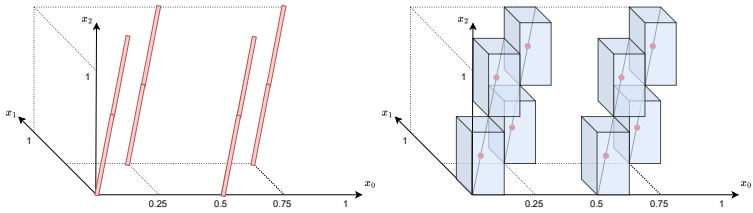
- Trajectories $\mathcal{B}_3^S$ uniformly distributed over 4 line segments.

Chaotic dynamics - sharp bounds and a minimal abstraction



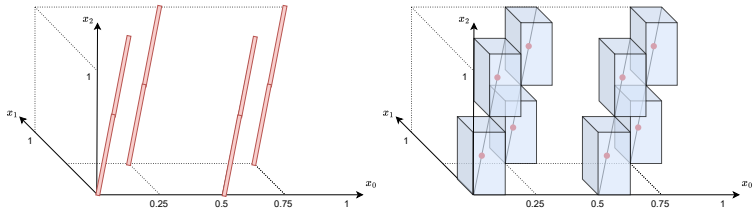
- Trajectories $\mathcal{B}_3^S$ uniformly distributed over 4 line segments.
- Optimal cover for $U(\mathcal{B}_3^S)$ (left figure).
  - Cut the lines into k equal pieces.

Chaotic dynamics - sharp bounds and a minimal abstraction



- Trajectories $\mathcal{B}_3^S$ uniformly distributed over 4 line segments.
- Optimal cover for $U(\mathcal{B}_3^S)$ (left figure).
  - Cut the lines into k equal pieces.
- Calculate $D_{cover}(R)$, with $R = \log(k2^{l-1})$.

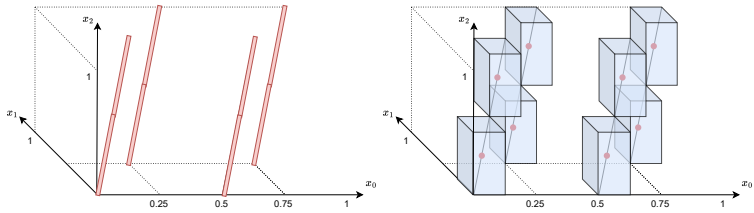
Chaotic dynamics - sharp bounds and a minimal abstraction



- Trajectories $\mathcal{B}_3^S$ uniformly distributed over 4 line segments.
- Optimal cover for $U(\mathcal{B}_3^S)$ (left figure).
  - Cut the lines into k equal pieces.
- Calculate $D_{cover}(R)$, with $R = \log(k2^{l-1})$.

- $\mathbb{E}[d(\xi, \Omega_A)|R] \geq D_{abs}(R) \geq D_{cover}(R)$.

Chaotic dynamics - sharp bounds and a minimal abstraction

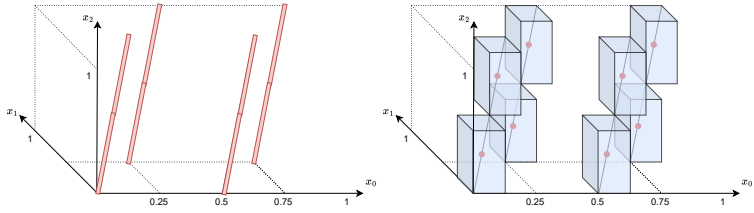


- Trajectories $\mathcal{B}_3^S$ uniformly distributed over 4 line segments.
- Optimal cover for $U(\mathcal{B}_3^S)$ (left figure).
  - Cut the lines into k equal pieces.
- Calculate $D_{cover}(R)$, with $R = \log(k2^{l-1})$.

- $\mathbb{E}[d(\xi, \Omega_A)|R] \geq D_{abs}(R) \geq D_{cover}(R)$.
- Here, we have $\mathbb{E}[d(\xi, \Omega_A)|R] = D_{cover}(R) \implies$

$$\boxed{\mathbb{E}[d(\xi, \Omega_A)|R] = D_{abs}(R)}$$

Chaotic dynamics - sharp bounds and a minimal abstraction



- Trajectories $\mathcal{B}_3^S$ uniformly distributed over 4 line segments.
- Optimal cover for $U(\mathcal{B}_3^S)$ (left figure).
  - Cut the lines into k equal pieces.
- Calculate $D_{cover}(R)$, with $R = \log(k2^{l-1})$.

- $\mathbb{E}[d(\xi, \Omega_A)|R] \geq D_{abs}(R) \geq D_{cover}(R)$.
- Here, we have $\mathbb{E}[d(\xi, \Omega_A)|R] = D_{cover}(R) \implies$

$$\boxed{\mathbb{E}[d(\xi, \Omega_A)|R] = D_{abs}(R)}$$

- Abstraction: partition $\mathcal{X}$ into $2^{l-1}k$ equal pieces.
- $\mathcal{B}_l^A$ covers $\mathcal{B}_l^S$ (right figure).

A 3D nonlinear system - abstraction conservatism?

System $x(k+1) = f(x(k))$, $x \in [-1,1]^3$:

- $f(x) = \begin{bmatrix} 0.9x_1 + 0.1\sin x_2 \\ 2x_2^3 - x_2 \\ 0.9x_3 + 0.1x_1x_2 \end{bmatrix}$

Abstraction:

- Uniform grid $N \times N \times N$.
- Not optimal!

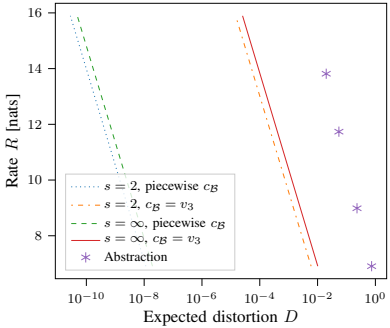
A 3D nonlinear system - abstraction conservatism?

System $x(k+1) = f(x(k))$, $x \in [-1,1]^3$:

- $f(x) = \begin{bmatrix} 0.9x_1 + 0.1\sin x_2 \\ 2x_2^3 - x_2 \\ 0.9x_3 + 0.1x_1x_2 \end{bmatrix}$

Abstraction:

- Uniform grid $N \times N \times N$.
- Not optimal!



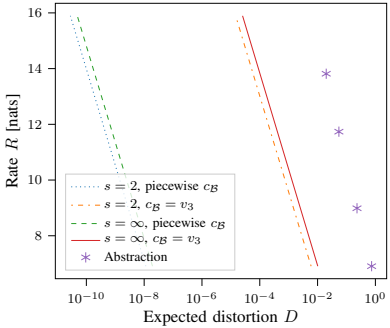
A 3D nonlinear system - abstraction conservatism?

System $x(k+1) = f(x(k))$, $x \in [-1,1]^3$:

- $f(x) = \begin{bmatrix} 0.9x_1 + 0.1\sin x_2 \\ 2x_2^3 - x_2 \\ 0.9x_3 + 0.1x_1x_2 \end{bmatrix}$

Abstraction:

- Uniform grid $N \times N \times N$.
- Not optimal!



Big gap:

- Is the bound bad or is the abstraction bad?
- How could we find a (nearly) minimal abstraction?

Conclusion

A quantitative theory on abstractions' size-accuracy tradeoff:

- Abstractions encode system trajectories in the form of covers in $\mathbb{R}^{nl}$.
- Rate-distortion theory for abstractions.
- Constructed minimal abstraction for doubling map.

Conclusion

A quantitative theory on abstractions' size-accuracy tradeoff:

- Abstractions encode system trajectories in the form of covers in $\mathbb{R}^{nl}$.
- Rate-distortion theory for abstractions.
- Constructed minimal abstraction for doubling map.

Fundamental scalability limits:

- Given system dynamics and desired accuracy, what is the smallest abstraction possible?
- Depends on dynamics complexity.
- Bounds are sharp in certain cases.
- Can be used as baseline.

Conclusion

A quantitative theory on abstractions' size-accuracy tradeoff:

- Abstractions encode system trajectories in the form of covers in $\mathbb{R}^{nl}$.
- Rate-distortion theory for abstractions.
- Constructed minimal abstraction for doubling map.

Fundamental scalability limits:

- Given system dynamics and desired accuracy, what is the smallest abstraction possible?
- Depends on dynamics complexity.
- Bounds are sharp in certain cases.
- Can be used as baseline.

Future work

How to construct minimal abstractions in a principled way?

References

This presentation is based on:

- G. Delimpaltadakis, and G. Gleizer (2025). "An Information Theory of Finite Abstractions and their Fundamental Scalability Limits." arXiv preprint arXiv:2512.03977. (under review)

Rate-distortion theory on measurable spaces:

- E. Riegler, H. Bölcskei, and G. Koliander (2018). "Rate-distortion theory for general sets and measures." in 2018 IEEE ISIT.
- E. Riegler, G. Koliander, and H. Bölcskei (2023). "Lossy compression of general random variables." Information and Inference: A Journal of the IMA, 2023.

For a literature survey on past efforts towards scalable abstractions (and what are they missing), check our paper's intro.